



Aakash

Medical | IIT-JEE | Foundations

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MM : 240

TYM for NEET-2025-26_Practice Test-01C

Time : 60 Min.

Topics Covered:**Physics:** Units and Measurements

Chemistry: Some Basic Concepts of Chemistry: Importance of chemistry, Nature of matter, Properties of matter and their measurement : Mass and weight, volume, density, temperature, Uncertainty in measurement, Scientific notation, Addition and subtraction, Multiplication and division, Significant figures, Dimensional analysis, Laws of chemical combination : Law of conservation of mass, Law of definite proportions, Law of multiple proportions, Gay lussac's law of gaseous volumes, Avogadro law, Dalton's atomic theory, Atomic and molecular masses : Atomic mass, Average atomic mass, Molecular mass, Formula mass, Mole concept, Molar mass, equivalent mass, Percentage composition, Empirical formula

Botany: Cell: The Unit of Life : Introduction, what is cell?, Cell theory, An overview of cell, prokaryotic cell structure, Gram staining, eukaryotic cell structure, Difference between prokaryotic and eukaryotic cell, Difference between plant cell and animal cell, Plasma membrane

Zoology: Structural Organisation in Animals: Animal tissues-I: Epithelial tissue, Connective tissue.

General Instructions :**Instructions:**

- (i) Use blue/black ballpoint pen only to darken the appropriate circle.
- (ii) Mark should be dark and should completely fill the circle.
- (iii) Dark only one circle for each entry.
- (iv) Dark the circle in the space provided only.
- (v) Rough work must not be done on the Answer sheet and do not use white-fluid or any other rubbing material on Answer sheet.
- (vi) Each question carries 4 marks. For every wrong response 1 mark shall be deducted from total score.

PHYSICS

1. If 9 divisions of main scale are equal to 10 divisions of Vernier scale and given that one main scale division is equal to 'a' units, then the least count of Vernier caliper is
 - (1) $\frac{a}{9}$
 - (2) $\frac{a}{8}$
 - (3) $\frac{a}{10}$
 - (4) $\frac{a}{11}$
2. If the dimensional formula of velocity is $[M^x L^y T^z]$, then $x + y + z$ is equal to
 - (1) Zero
 - (2) 1
 - (3) 2
 - (4) 3
3. The equivalent capacitance for two capacitors connected in parallel is given by $C_{eq} = C_1 + C_2$. If $C_1 = (10 \pm 1)\mu F$ and $C_2 = (20 \pm 1)\mu F$, then the equivalent capacitance in parallel can be written as
 - (1) $(30 \pm 1)\mu F$
 - (2) $(30 \pm 2)\mu F$
 - (3) $(30 \pm 1.5)\mu F$
 - (4) $(30 \pm 3)\mu F$
4. Which of the following physical quantities have the same dimensional formula as work?
 - (1) Power
 - (2) Energy
 - (3) Force
 - (4) Momentum

5. If percentage error in measurement of side of a cube is 2% and in its mass is 3%. The maximum percentage error in measurement of its density is
- 5%
 - 7%
 - 9%
 - 2%
6. Two measured quantities are given as $A = (2.0 \text{ m} \pm 0.1) \text{ m}$ and $B = (2.0 \text{ m} \pm 0.1) \text{ m}$. The correct value of $(AB)^{1/2}$ will be
- $2.0 \text{ m} \pm 0.1 \text{ m}$
 - $2.0 \text{ m} \pm 0.2 \text{ m}$
 - $1.0 \text{ m} \pm 0.1 \text{ m}$
 - $1.0 \text{ m} \pm 0.2 \text{ m}$
7. If velocity of a body is given by $v = at^{3/2}$, then dimension of a should be (t is time)
- $[LT^{5/2}]$
 - $[L^{-1}T^{5/2}]$
 - $[LT^{-5/2}]$
 - $[LT^{-3/2}]$
8. If the unit of mass, length and time are doubled, unit of force will be
- Doubled
 - Tripled
 - Remains same
 - Quadrupled
9. The number of significant figures in measurement 0.0400200m is
- 4
 - 6
 - 2
 - 3
10. The value of 30° in radian is
- $\frac{\pi}{2}$
 - $\frac{\pi}{6}$
 - $\frac{\pi}{4}$
 - $\frac{\pi}{5}$
11. The most precise reading of length of an object among the following is
- 20 m
 - 20.01 m
 - 20.0 m
 - $20 \times 10^0 \text{ m}$
12. The value of $100 + 0.423$ upto correct decimal place is
- 100.423
 - 100.42
 - 100
 - 100.4
13. If $Z = P + Q$, then maximum percentage error in the measurement of Z will be
- $\left(\frac{\Delta P}{P} + \frac{\Delta Q}{Q}\right) \times 100 \%$
 - $\left(\frac{\Delta P}{P+Q} + \frac{\Delta Q}{P+Q}\right) \times 100 \%$
 - $\left(\frac{\Delta P}{Q} + \frac{\Delta Q}{P}\right) \times 100 \%$
 - $\left(\frac{\Delta P}{P+Q} - \frac{\Delta Q}{P+Q}\right) \times 100 \%$
14. If the unit of power is 1 kilo watt, the length is 100 m and time 1 s, then the unit of mass will be
- 10 kg
 - 10^{-1} kg
 - 100 kg
 - 10^{-2} kg
15. Random error can be minimised by
- Taking less number of readings
 - Improving experimental techniques
 - Taking large number of readings
 - Both (1) and (2)

16. Identify the example of pure substance among the following
- (1) Milk
 - (2) Air
 - (3) Sugar solution
 - (4) Carbon dioxide
17. Law of conservation of mass is not valid for
- (1) Decomposition reactions
 - (2) Neutralization reactions
 - (3) Nuclear reactions
 - (4) Combustion reactions
18. 0°F temperature is equal to
- (1) -15.55°C
 - (2) -17.77°C
 - (3) -32°C
 - (4) -18°C
19. The prefix pico stands for
- (1) 10^{-12}
 - (2) 10^{-15}
 - (3) 10^{-9}
 - (4) 10^{-6}
20. The number 232.508 in scientific notation can be written as
- (1) 2.32508×10^2
 - (2) 232508×10^{-3}
 - (3) 0.232508×10^3
 - (4) 23.2508×10^1
21. Number of significant figures present in 0.0033 is
- (1) 2
 - (2) 4
 - (3) 5
 - (4) Infinite
22. The empirical formula of an oxide of carbon which has 52.94% carbon and 47.06% oxygen by mass is
- (1) CO
 - (2) CO₂
 - (3) C₃O₂
 - (4) C₃O₄
23. The mass percentage of oxygen in H₂SO₄ is
- (1) 15.23
 - (2) 65.31
 - (3) 17.15
 - (4) 23.24
24. Number of S-atoms present in 0.5 g molecules of S₈ is
- (1) 1.98×10^{23} atoms
 - (2) 2.14×10^{23} atoms
 - (3) 2.41×10^{24} atoms
 - (4) 1.25×10^{24} atoms
25. Which has the maximum number of molecules among the following?
- (1) 36 g H₂O(l)
 - (2) 64 g CH₄(g)
 - (3) 44 g CO₂(g)
 - (4) 10 g H₂(g)
26. Among the following, maximum number of atoms are present in
- (1) 14 g N₂
 - (2) 64 g SO₂
 - (3) 88 g CO₂
 - (4) 8 g H₂
27. If the mass ratio of He to N₂ in a container is 2 : 1 then the ratio of their moles will be
- (1) 2 : 1
 - (2) 4 : 1
 - (3) 7 : 1
 - (4) 14 : 1
28. The SI unit of thermodynamic temperature and luminous intensity respectively are
- (1) Kelvin and Ampere
 - (2) Fahrenheit and Candela
 - (3) Kelvin and Candela
 - (4) $^{\circ}\text{C}$ and Ampere

29. For which of the following compounds empirical formula and molecular formula are same?

- (1) CH_3COOH
- (2) $\text{HOCH}_2\text{CH}_2\text{OH}$
- (3) C_6H_6
- (4) CH_3CHO

30. Mass of one amu in kg is

- (1) 1.66×10^{-27}
- (2) 6.022×10^{-27}
- (3) 6.022×10^{-23}
- (4) 1.66×10^{-24}

BOTANY

31. When glycocalyx is thick and tough, it is called

- (1) Capsule
- (2) Slime layer
- (3) Cell wall
- (4) Cell membrane

32. Bacteria reproduce mainly by

- (1) Endospore formation
- (2) Fission
- (3) Sexual means
- (4) Spore formation

33. Select the inclusion body, found in prokaryotes

- (1) Polysome
- (2) Mesosome
- (3) Glycogen granule
- (4) Chromatophore

34. The function of polysome is to

- (1) Translate the mRNA into protein
- (2) Synthesise lipids in eukaryotic cells
- (3) Replicate DNA during cell division
- (4) Store various hydrolytic enzymes

35. In human beings, the membrane of erythrocytes has approximately ____% of proteins and ____ % of lipids respectively

- (1) 40, 52
- (2) 52, 40
- (3) 40, 40
- (4) 60, 40

36. All of the given functions are performed due to the fluid nature of cell membrane, **except**

- (1) Cell growth
- (2) Endocytosis
- (3) Formation of intercellular junction
- (4) Protection of cells from attack of pathogens

37. In addition to phospholipids, cell membrane contains all of the following molecules, **except**

- (1) Cellulose
- (2) Protein
- (3) Cholesterol
- (4) Carbohydrate

38. Select the **incorrect** statement w.r.t. plasmid.

- (1) It is larger than genomic DNA
- (2) It is circular DNA outside genomic DNA
- (3) It confers certain unique phenotypic characters to bacteria
- (4) It can provide resistance against antibiotics to bacteria

39. A bacterial flagellum is composed of

- (1) Inclusion bodies, filament and hook
- (2) Filament, hook and basal body
- (3) Filament, cell wall and cell membrane
- (4) Filament, mesosomes and basal body

40. _____ are involved in respiration in prokaryotes. Choose the **correct** option to fill in the blank
- (1) Ribosomes
 - (2) Polysomes
 - (3) Mesosomes
 - (4) Lysosomes
41. Which layer of the cell envelope prevents the bacterium from bursting or collapsing?
- (1) Plasma membrane
 - (2) Glycocalyx
 - (3) Cell wall
 - (4) Capsule
42. The cell organelle or structure common in both animal and higher plant cell is
- (1) Centriole
 - (2) Large central vacuole
 - (3) Ribosome
 - (4) Plastid
43. Which of the following is a site of protein synthesis?
- (1) Lysosome
 - (2) Ribosome
 - (3) Centrosome
 - (4) Smooth endoplasmic reticulum
44. Which one of the following structures allows bacteria to hide from host's immune system?
- (1) Capsule
 - (2) Slime layer
 - (3) Cell wall
 - (4) Flagella
45. The concept of *Omnis cellula-e cellula* given by Rudolf Virchow suggests that
- (1) All living organisms are composed of cells.
 - (2) All living organisms are composed of products of cells.
 - (3) All cells arise from pre-existing cell.
 - (4) Activities of an organism are the outcome of sum total of activities and interactions of its constituent cells.

ZOOLOGY

46. The epithelial tissue lining the inner surface of oviducts is
- (1) Simple cuboidal
 - (2) Simple squamous
 - (3) Pseudostratified epithelium
 - (4) Simple ciliated columnar
47. In epithelial tissue
- (1) Cells are compactly packed with large amounts of intercellular matrix
 - (2) A free surface is present which faces either a body fluid or the outside environment
 - (3) The tissue provides a protective covering or lining for all parts of the body
 - (4) The cells are always loosely packed with little intercellular matrix
48. Alveoli of lungs and blood vessels are lined by
- (1) Columnar epithelium
 - (2) Squamous epithelium
 - (3) Cuboidal epithelium
 - (4) Pseudostratified epithelium
49. Which of the following tissues main function is to provide protection against chemical and mechanical stresses?
- (1) Simple epithelium
 - (2) Compound tissue
 - (3) Neural tissue
 - (4) Adipose tissue
50. Blood is a fluid connective tissue and is in direct contact with which type of epithelial tissue during circulation?
- (1) Columnar epithelium
 - (2) Cuboidal epithelium
 - (3) Simple squamous epithelium
 - (4) Stratified squamous epithelium
51. The brush border cells are required to increase the surface area for absorption. In humans, these cells are present on inner lining of
- (1) Bronchioles and fallopian tubes
 - (2) Fallopian tubes and pancreatic ducts
 - (3) Eustachian tube and bronchioles
 - (4) Small intestine and PCT of nephrons

52. Which of the following tissues forms covering over almost all the body organs?
- Nervous tissue
 - Epithelial tissue
 - Connective tissue
 - Muscular tissue
53. Read the following statements and choose the **correct** option.
Statement A : Some of the squamous and cuboidal cells get specialised for secretion, called glandular epithelium.
Statement B : Salivary gland is an example of multicellular glandular epithelium consisting of cluster of cells.
- Both statements A and B are incorrect
 - Both statements A and B are correct
 - Statement A is correct and B is incorrect
 - Statement B is correct and A is incorrect
54. Choose the **odd** one w.r.t. specialized connective tissues.
- Bone
 - Blood
 - Collagen
 - Cartilage
55. Consider the following connective tissue.
a. Bone
b. Cartilage
c. Areolar tissue
d. Tendon
e. Ligaments
How many among the above can have collagen fibres?
- Only one
 - Only two
 - Only four
 - All five
56. Which type of simple epithelial tissue is found in the lining of intestine facing the lumen?
- Squamous epithelium
 - Cuboidal epithelium
 - Columnar epithelium
 - Ciliated epithelium
57. Single layer of flattened polygonal cells with irregular boundaries form
- Squamous epithelium
 - Cuboidal epithelium
 - Columnar epithelium
 - Ciliated columnar epithelium
58. Tissue whose cells are specialised to store fat and is located mainly beneath the skin is
- Areolar tissue
 - Adipose tissue
 - Tendon
 - Ligament
59. Which amongst the following is a special function of connective tissue?
- To link and support other tissues/organs of the body
 - It exerts the greatest control over the body's responsiveness to changing conditions
 - To play an active role in all the movements of the body
 - Moves the body to adjust to the changes in the environment
60. Find the **odd** one out w.r.t. cells of areolar connective tissue.
- Mast cells
 - Macrophages
 - Fibroblasts
 - Goblet cells